

Turbulence modulation via dilute surfactant additives in experimental Taylor–Green flow

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Abstract here.

I. INTRODUCTION

Since the discovery that trace amounts of dissolved polymer could lead to drag reduction in pipe flows [1], the use of additives for drag reduction in wall bounded flows has become established. Besides polymer solutions, it has since been discovered that self-assembling surfactant solutions share similar effects [2], owing to the fact that dilute surfactant solutions can form wormlike micelles whose relaxation dynamics mimic that of polymer chains while resisting mechanical degradation. Beyond the applications of drag reduction, it has shown that viscoelastic additives are known to modulate turbulence more broadly, reshaping the energy spectrum and producing an elastic range governed by additive elasticity rather than inertial at the small scales [3, 4], with surfactant-specific micellar structures shown to alter vortex shedding and turbulence statistics directly [5].

In spite of academic and commercial interest, comparatively few studies address the behavior of dilute surfactant solutions in the context of three-dimensional bulk turbulence, with most of the literature focused on wall-bounded or quasi-two-dimensional flows. This is in part related to the fact that the rheology of very dilute surfactant solutions remains difficult to pin down, with reported values of viscosity and relaxation time of CTAC/NaSal solutions (such as the considered in the present experiment) varying considerably between studies [6, 7]. In spite of the diverging measurements and reports that some of the dilute solutions behave as a Newtonian fluid under rheological measurements [6], such solutions are known to produce measurable drag reduction [8], which further reinforces the need to analyze such solutions more in depth.

Motivated by this gap, here we present an experimental analysis of the turbulence generated by a four-rotor, Taylor–Green-like experiment using dilute CTAC solutions, in an attempt to characterize the modulation of turbulence caused by the addition of the surfactant across a wide range of Reynolds numbers for flows driven by inertial forcing.

II. EXPERIMENTAL DEVICE

In the present study, we employ an experimental device designed to produce Taylor–Green-like forced turbulence on a laboratory setting. The apparatus, with schematic shown on figure 1, consists of a water tank

with four rotors which are driven by a gear system and rotate in opposing directions in order to produce turbulence in the region between them (hereinafter simply denominated bulk), which is the target of our studies. Spatially-resolved measurements are obtained via 2D particle tracking (briefly described in section II A) via a traditional laser-sheet and high-speed camera combination, which are positioned as also shown in figure 1 for the case of bulk imaging.

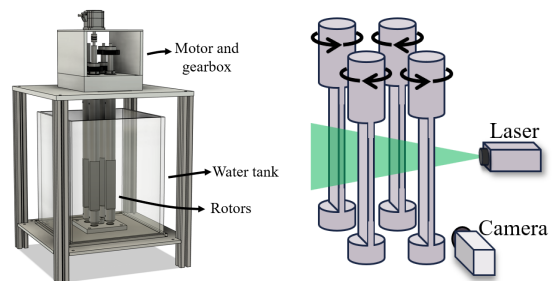


FIG. 1. CAD of the experimental device (left) and schematic showing normal orientation of camera and laser sheet, as well as direction of rotation of the rotors.

Aside from the imaging of the bulk region, it is also possible to change the relative orientations of the laser sheet and camera, and use a mirror to allow for top-down imaging of the region in the vicinity of the blades, as shown in the schematic of figure 2.

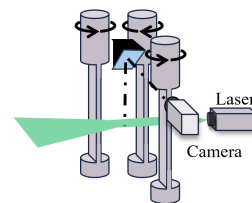


FIG. 2. Schematic of the altered orientation of camera and laser using mirror for imaging of the space between rotors. One of the rotors is omitted for clarity.

As shown in the schematics of figures 1 and 2, the rotors themselves are shaped as blades with rectangular cross-section. While the design of the experimental device allows for the geometry of the rotors to be changed between cylinders and the carved blades, the use of the latter allows us to isolate inertial forcing in the driving of the flow, and therefore ignore boundary-layer effects which would be present in the case of viscous driving

which happens when cylindrical rotors are used. Additionally, the rotation direction is such that mean stretching of the flow happens in the out-of-plane component of the camera image (when imaging the bulk region in the vertical plane), and is accompanied by the formation of secondary vortices as shown in previous numerical studies [9]. For simplicity, the carved rotors will hereinafter simply be denominated blades.

While not shown in the schematics, the experimental device is also encased in thermal insulation and connected to an air-processor system. By controlling the temperature of the surrounding air, the water temperature was maintained at approximately 26 °C over the course of the experiments.

A. Data measurement and reduction

As previously mentioned, processing of the obtained images to calculate the velocity fields is conducted via a particle tracking velocimetry (PTV) code developed in-house. Although the details of the algorithm are out of the scope of the current document, the code is built as a collection of previously established methods. For the detection of tracer particles from the images, we employ a variation of the Iterative Particle Reconstruction (IPR) method, which is commonly employed for volumetric particle tracking as part of the Shake-the-Box method [10]. The tracking of detected particle positions between the video frames is carried out by a variation of the Polar Coordinate-System Similarity (PCSS) method [11].

Before the calculation of the final velocity vectors through finite differencing, the trajectories are filtered according to a Wiener Filter [12] which is calculated from the spectra of position of the obtained trajectories. The filtering is done in order to avoid the effect of the noise floor in the calculation of the velocities. (need to revisit this) (also need to mention the interpolation that is done before the spatial average) **also supplement this with the explanation of the uncertainty calculation and all.**

B. Surfactant solutions

The surfactant used in the current study was the cationic surfactant cetrimonium chloride (CTAC), with the addition of sodium salicylate on a molar ratio of approximately 4:1 to provide counterions. The solution is prepared by mixing the adequate amounts of surfactant and salt with tap water and then agitating in a magnetic stirrer over a period of at least 12 hours, after which the resulting solution is added to the experimental device. For simplicity, the surfactant and salt solution are hereinafter denominated simply as the CTAC solution, with the solution concentration corresponding exclusively to the concentration of the surfactant counterpart.

For the analysis of the effect of solution concentration on the results, three CTAC solutions of concentrations of

33 ppm, 50 ppm, and 100 ppm were prepared. As previously mentioned, rheological measurements of such dilute solutions have historically yielded diverging results [6, 7], meaning that to the best of the author's knowledge, there is no precise measurement of the properties such as the relaxation time of micellar structures under these conditions.

Moreover, the molar ratio of 4:1 chosen for the current study differs from that of the aforementioned measurements, as well as that of drag reduction studies [8]. Although numerical studies have reported that such high molar ratios can lead to the formation of branched structures and loss of viscoelasticity [13], such claims have been made for solutions of much higher concentrations. In the experience of the authors, this effect is not as pronounced in the case of dilute solutions, and the use of a high molar ratio has been chosen to favor the formation of longer micellar structures.

In spite of these divergences, it can be expected from the aforementioned studies that this relaxation time should be in the order of 10^{-1} seconds. Since the relaxation time cannot be precisely known beforehand or independently controlled over a wide range, a sweep over the Weissenberg **you need to define what is the weissenberg number** **bro** number is instead achieved by changing the Reynolds number of the generated turbulence. For each of the prepared solutions, the rotation rate of the blades is changed 2^{-3} to 2^4 Hz in powers of two. Exclusively for the 100 ppm solution **say how the rotation rate was reduced further**. Although it is debatable whether the flow generated in the case of the lower rotation rates can be denominated turbulent, it remains chaotic and unsteady, so such cases are still useful in the determination of the effect of surfactant addition on such unsteady but low Reynolds number types of flows.

Note that although it can be expected that the precise value of τ_s should change with the concentration of the solutions, the results of the aforementioned studies indicate that this change is not necessarily monotonic. Nevertheless, the same results indicate that the variations in relaxation time should be small enough for the precise value to remain in the same order of magnitude. For this reason, we take the representative value of $\tau_s = 0.1$ s as the boundary at which $W_i \approx 1$ for the three solutions. With this consideration, the range of Weissenberg numbers are obtainable in the current experiment range from $W_i = \mathcal{O}(10^{-2})$ to $W_i = \mathcal{O}(10^2)$, allowing us to see the transitions to the attenuated regime regardless of small changes in the precise value of relaxation time.

The next section describes briefly the characterization of the turbulence generated in the current experimental device for the case of Newtonian fluid.

III. TURBULENCE CHARACTERIZATION

Considering the unique nature of the current experimental apparatus, we conduct a characterization of the

turbulence generated in the case of Newtonian flow as a function of the rotation rate of the blades. Although the generated turbulence is anisotropic, turbulent statistics become approximately homogeneous in the vicinity of the stagnation point of the mean flow in the center of the experiment. Owing to this, the statistics are computed as spatiotemporal averages over the duration of the experiments. Figure 3 displays the velocity fluctuation RMS as a function of the blade rotation speed, demonstrating that there is an approximately linear trend, with the ratio between the two components remaining somewhat constant as well. Note that the turbulent intensity is more pronounced in the horizontal direction, consistent with the primary forcing direction of the blades.

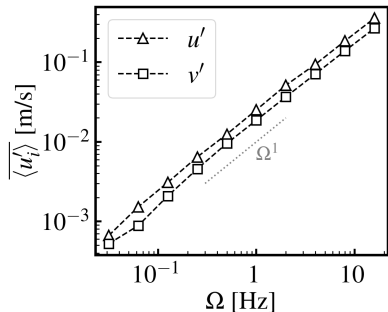


FIG. 3. Spatially averaged velocity fluctuation RMS in the bulk as a function of rotation rate of the agitator blades.

Although direct measurement of the dissipation range is not feasible under current experimental conditions, the turbulent dissipation can be estimated through the third-order structure function, as shown in figure 4. Note that although the generated turbulence is anisotropic, the results of figure 4 indicate that the dissipation rate is approximately independent of the measurement direction, which helps to validate the obtained results **is this correct?**. In all the plots, the increase in the value of the structure function as the separation distance reaches the smallest values is caused by the finite depth-of-correlation and the nature of planar imaging, as well as the fact that no interpolation of the instantaneous velocity fields is used for the calculation of these statistics.

For the cases of high rotation rate, the appearance of a horizontal region indicates that an inertial range is well-defined, and the dissipation rate can be taken as the average across this region. For the lower rotation rates, the Reynolds number is too low for the appearance of a defined inertial range, and the extraction of a representative value of dissipation is more difficult. For consistency between the cases, we consider the average value of the normalized third-order structure function between the separations of $r = 3$ mm and $r = 20$ mm as the dissipation value. These representative values are represented as the dotted lines of figure 4.

From these results, we can then obtain the approximate Kolmogorov lengthscale and timescale for each of the cases, as shown in figure 5. From these results, the es-

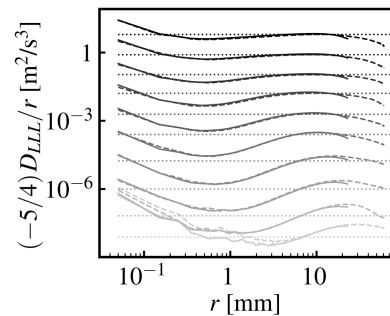


FIG. 4. Normalized third-order structure function for different rotation rates. Darker lines represent a higher rotation rate.

timated Kolmogorov lengthscale provides us with an additional clue as to why it is not possible to directly measure the dissipation range, considering that these scales become smaller than the depth-of-correlation of the current imaging system for most rotation rates. Additionally, wide range of the estimated timescales indicate that a wide range of Weissenberg numbers should be obtained regardless of the fact that the relaxation time of the surfactant solutions is not precisely known.

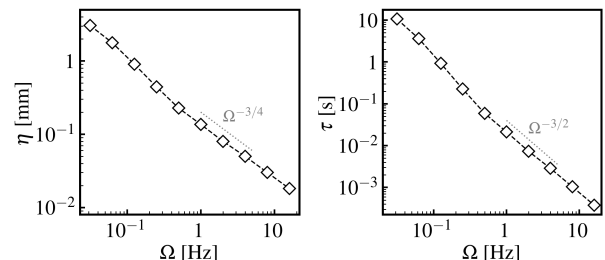


FIG. 5. Kolmogorov lengthscale and timescale as a function of rotation rate of the agitator blades.

IV. RESULTS

The analysis of turbulence modulation by surfactant addition begins with the turbulent intensity at the bulk region, which is described here by the spatially averaged RMS of the velocity fluctuations. Since we conduct planar measurements, this magnitude is calculated from the two measured components, by defining $q'_{2D} = \sqrt{\langle u'^2 \rangle + \langle v'^2 \rangle} = \sqrt{u'^2_{rms} + v'^2_{rms}}$, which is hereinafter referred to as q' for simplicity. The results are then spatially averaged across the measurement region, since these statistics are approximately homogeneous in this area as previously discussed. Note that although the value of q' does not contain the out-of-plane component of the turbulent kinetic energy, its magnitude can be expected to be similar to the currently measured horizontal component of the velocity fluctuations. With this

in mind, the results of q' are interpreted as an indication of the trends of turbulent kinetic energy in the bulk region. Aiming to observe the modulation of such a quantity, figure 6 then presents the ratio of the spatially averaged velocity fluctuation RMS between the surfactant and Newtonian cases.

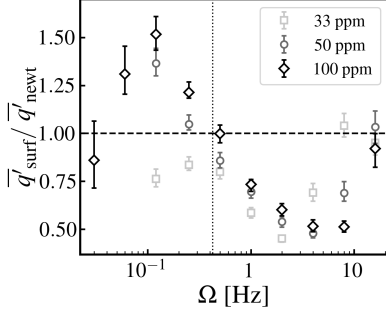


FIG. 6. Ratio of spatially averaged velocity fluctuation RMS between the surfactant and Newtonian cases. Vertical dashed line represents the rotation rate for which $W_i \approx 1$ considering a relaxation time of $\tau_s = 0.1$ s.

According to the classical picture of turbulence modulation via the addition of surfactants and polymers, eddies with a turnover time shorter than the relaxation time of the viscoelastic solutions should be attenuated, and the turbulence in the regime where $W_i < 1$ should remain approximately unchanged. The results shown in figure 6 however, display a more complex picture. It can be seen that although turbulent energy is indeed attenuated once the Weissenberg number crosses the frontier of unity around all cases, the turbulence also appears to be intensified for the 50 ppm and 100 ppm solutions at low rotation rates. Furthermore, once the rotation rate is increased past a certain threshold, the attenuation is lost and the solutions appear to return to the Newtonian behavior.

To help us understand the reason behind these phenomena, We can start by looking at the effect of surfactant addition on mean flow intensity, which can be defined as the average flow velocity at a given region around the stagnation point. Considering the symmetric nature of the experimental apparatus, as well as the forcing with respect to the right and left regions of the measurement domain, we can enforce this symmetry on the mean flow to improve convergence of the results. This is done by defining a mirror coordinate system as $\tilde{\mathbf{x}} \equiv (2x_c - x, y)$, with $x = x_c$ corresponding to the symmetry axis, or stagnation point. This then allows us to write a symmetry-enforced mean flow field component, defined as:

$$U^\dagger(\mathbf{x}) = \frac{1}{2} [U(\mathbf{x}) - U(\tilde{\mathbf{x}})]. \quad (1)$$

We can then compute the spatial average of this quantity to then obtain a representative intensity of the mean flow. Since the field is anti-symmetric however, this spatial average is computed only on the left-hand side of

the point (such that $x < x_c$), resulting in the quantity $\bar{U}_L^\dagger$. Note that this approach is chosen (in opposition to taking the spatial average of the velocity magnitude) because it allows us to preserve the flow direction, and thus take into account flow reversal as well should it happen. Moreover, since the defined quantity has the significance of “spatially averaged mean flow” in the bulk region, it will hereinafter simply be denominated as $\bar{U}$ for visual clarity. With this in mind, figure 7 presents the ratio of mean flow intensity between the surfactant and Newtonian cases.

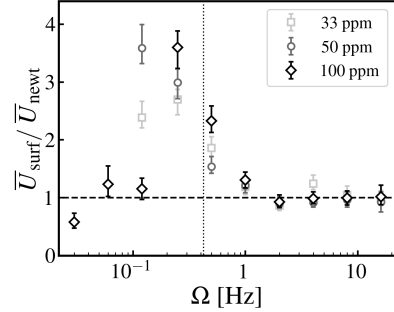


FIG. 7. Ratio of spatially averaged horizontal velocity magnitude between the surfactant and Newtonian cases. Vertical dashed line represents $W_i \approx 1$.

From the plot of figure 7, it can be seen that for rotation rates of approximately 1 Hz or above, the addition of surfactant has almost no effect on the mean flow, irrespective of the concentration. It can be stated that such would be the expected behavior for turbulence which is inertially driven, since the role of surfactant additives is traditionally considered as to attenuate the small scales of the flow. Since the driving rotors act on the large scales, which even at the slowest rotation rate are larger than the Kolmogorov lengthscale by a factor of at least one decade (ref. fig. 5), it can be expected that such large scales should remain mostly unaffected. Nevertheless, the results from figure 7 indicate that reducing the rotation rate leads to an increase of up to 4 times on the mean flow intensity for the regime where $\Omega \lesssim 1$ Hz. Additionally, this behavior appears to be independent of the frontier of $W_i \approx 1$, with the peak in mean flow intensification occurring well below the value of unity of the Weissenberg number, after which the amplification effect appear to cease once the rotation rate is further decreased.

To investigate the cause behind this increase in the mean flow for the lower rotation rates, figure 8 presents the value of the same spatially averaged horizontal component of the mean velocity, but as a function of the phase of the blade (ϕ). Similarly to the previously defined quantities, we introduce $U_\phi^\dagger(\mathbf{x}, \phi) = \frac{1}{2} [U_\phi(\mathbf{x}, \phi) - U_\phi(\tilde{\mathbf{x}}, \phi)]$ corresponding to the symmetry-enforced version of the phase-locked component of the mean flow U_ϕ . Again, for visual clarity, this quantity is

hereinafter simply denominated as $\overline{U}_\phi$ after the spatial averaging over the left side of the observed domain is taken.

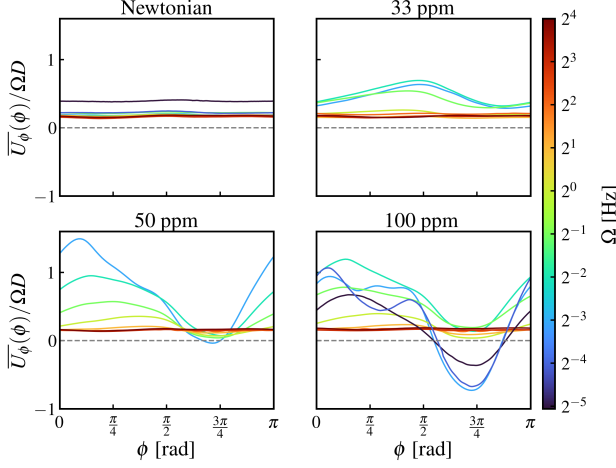


FIG. 8. Spatially averaged horizontal mean flow component as a function of blade phase for each of the tested cases. Positive value refers to inward motion (towards the stagnation point), while a negative value indicates flow reversal.

From these plots, it is possible to see that the overall dependence of the mean flow with the phase of the blade gets amplified as the concentration of the solution is increased and the rotation rate is decreased, reaching a peak after which the influence starts to decrease once the rotation rate becomes too slow. In the cases of the rotation rate of 2^{-4} and 2^{-3} Hz in the 100 ppm solution, strong reversal of the flow direction also occurs during part of each cycle, even though the blades are always rotating in the direction of pushing the fluid towards the center of the bulk region.

Figure K presents the amplitude of these curves, normalized by their respective rotation rates. It is possible to observe that while the Newtonian case displays approximately constant normalized amplitude, the surfactant cases display the presence of a peak in the lower rotation rates. As the rotation rate is increased, the amplification of the phase-locked flow amplitude falls, until a small attenuation is seen before the curves join that of the Newtonian flow at the highest rotation rate. While the peak was not observed for the 50 ppm case due to lack of measurement points, it is expected that this peak should lie between the 33 and 50 ppm curves.

The behavior observed in figures C and K is indicative that there is a frequency of maximum amplification of the mean velocity field, and that the classical attenuation of small turbulent scales is not the only phenomenon at play in the current experiment. Considering that the periodic behavior of the mean flow is governed by the largest flow scales, which have frequencies much lower than the expected relaxation time of micellar structures, it can be hypothesized that there is a second type of the interaction between the rotating blade and large-scale structures

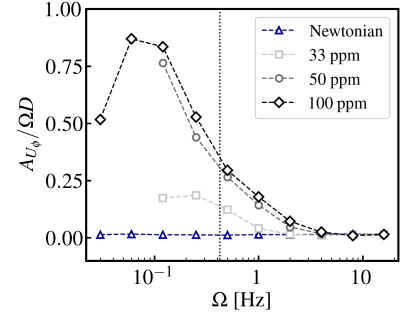


FIG. 9. Normalized amplitude of the spatially averaged, phase-locked horizontal component of the mean flow as a function of rotation rate.

created by the surfactant additives.

In order to analyze the flow structures which are present in the flow, we can rely on the flow fields measured in the horizontal plane and in the vicinity of the blades, with the technique shown previously shown in figure 1(c). Considering the limited viewing area however, the line of sight of the camera is rotated horizontally by a few degrees such that the imaging region lands between two of the blades. The schematic displayed in figure K shows the actual measurement area, displayed as the square which overlaps part of the two leftmost blades, and containing some of the measured velocity vectors. For context the bulk region, as well as the laser sheet position and camera viewing direction that are used for the imaging of the bulk region are also shown. Moreover, the points defined as the leading and trailing edges of each blade profile are also highlighted. Note that for clarity reasons, only a portion of the measured vectors are displayed.

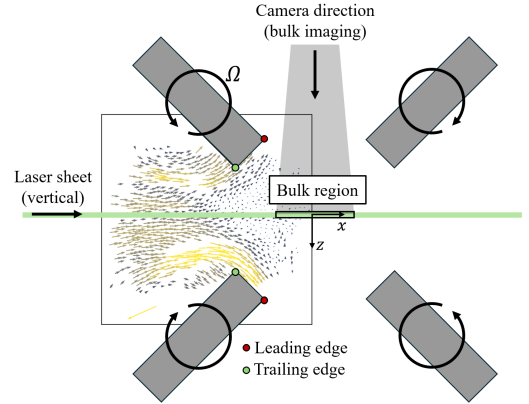


FIG. 10. Schematic of the viewing area in the horizontal plane with respect to the position of the blades, as well as laser sheet and camera directions during bulk imaging.

With this schematic in mind, figure E displays the evolution of the flow field over the course of half a rotation period divided in four snapshots, for the Newtonian and 50 ppm cases at the rotation rate of 0.125 Hz. For the

Newtonian flow case, displayed on the left column, no particularly noteworthy flow structures are visible. The highest flow velocity is seen in the vicinity of the blades (particularly behind the trailing edge), and a gradual attenuation of the overall velocity magnitude is seen towards the bulk region. When the surfactant is added however, the picture becomes different.

In the rightmost column of figure D, at the starting position with both blades into view, the flow is most constrained, and the motion of the blades is accompanied by broader (when compared to the Newtonian flow) higher velocity regions which extend from the leading edge to the trailing edge direction. As the two blades begin to separate in the second snapshot, these two faster moving regions transform into a more developed shear-layer structure, eventually developing into multiple streaks once the blades are parallel to each other (in the third snapshot). Such streaks act like jets which impinge at the bulk region, leading to the perceived increase in mean flow intensity. Looking back at the first snapshot, it can also be seen that these jets disappear quite quickly over the span of 45 degrees of rotation. Combined with the elastic nature of such non-Newtonian fluids, it can be expected that this compressive motion in the x direction, induced by the jets, which is then accompanied by a relaxation of the micellar structures once the jets have disappeared, can lead to the observed reversal of the flow direction under certain conditions.

The streak-like structures shown in figure C show resemblance to the elongated structures observed in drag-reduction studies in channel flow using solutions of similar concentrations [14]. In this study, the authors argue that the formation of shear-induced-structures (SIS) are a key part of the drag-reducing capability of surfactant, and the PIV measurements of wall-parallel planes of the drag reducing solutions show similar elongated structures in the vicinity of the walls. In our current experiment, there exists a region of mean shear between the blades due to the velocity gradient. It can be hypothesized that this shear, coupled with the relatively low turbulent intensity of the the slower cases leads to favorable conditions for the formation of SIS, which we see as the jet-like structures of figure C. It can then be conjectured that for the slower rotation rates (below 1 Hz), the interaction between the rotation rate of the blades causes the acceleration of these structures which leads to the increase in the mean flow. As the shear rate increases further however, their formation is no longer favored and the mean flow intensity returns that of the Newtonian fluid. Moreover, the fact that reducing the rotation rate even further reduces the amplitude of velocity fluctuations (such as shown in figure B for the slowest case of the 100 ppm solution) is may be and indicative of a mismatch in temporal scales between the rotation rate of the blades and the elasticity of the large structures, which decreases the amplitude of the oscillations.

With the effect of the surfactant addition in the mean flow in mind, we can also analyze what happens to the

turbulent intensity in a different manner. Since the previous analyses have indicated the dependence of the mean flow with the blade phase, it is important to take this into account when computing the change in turbulent intensity that occurs with the addition of the surfactant. In order to account for these periodic oscillations the, phase-dependent mean flow (shown in figure B), is used when calculating the velocity fluctuations in this stage of the analyses. The results from each of the surfactant solutions are then contrasted against the Newtonian flow, and the results as a function of the blade rotation speed are shown in figure E. These results then become more representative of the actual effect of the surfactant on the turbulent intensity inside the experimental device, allowing us to discuss some trends.

From this plot we can see that the behavior of of turbulent intensity shows different trends depending on the ranges of rotation rate. We start by analyzing the trend which is observed in the middle region of the plot, for rotation rates between approximately 0.5 Hz and 2 Hz or 8 Hz depending on the concentration of the solution. This is the region which is most representative of the traditional turbulence modulation under the influence of polymers or surfactants, and is characterized by the attenuation of turbulence. As previously discussed, it is a well known phenomenon that turbulent scales with timescale faster than the relaxation time of the surfactant micelles are attenuated. The fact that the velocity fluctuation RMS starts to decrease due to the surfactant addition for rotation rates higher than 0.5 Hz indicates that the Weissenberg number surpasses the unitary value at approximately this point. Moreover, since the point at which this attenuation begins for the three solutions is approximately the same (begin 0.25 and 0.5 Hz), we can conclude that the relaxation time of the three surfactant solutions is approximately the same as well. Referencing turbulence characterization studies from Section ???, we conclude that this relaxation time is in the vicinity of 0.08 seconds, which is in agreement with the expected values from the studies of Zhang and Kawaguchi **I think this part needs to be refactored.**

As the Reynolds number is further increased, we see another change in trend of turbulent intensity: the attenuation effect is diminished, and the turbulent intensity progressively returns to the value of the Newtonian fluid once again. This behavior is analogous to what has been observed in studies of drag reduction in channel flow (Motozawa 2011, Wei 2011), where a critical Reynolds number exists past which the drag reduction effectiveness sharply declines and the flow returns to the Newtonian state. In such studies, the authors argue that while presence of some shear is essential for the formation of shear-induced-structures which play a key role in the drag reduction, excessive shear leads to the breakup of such structures and inhibit their formation, which undermines the drag reduction effectiveness.

While the current experimental arrangement differs from the case of a channel flow, analogous conclusions

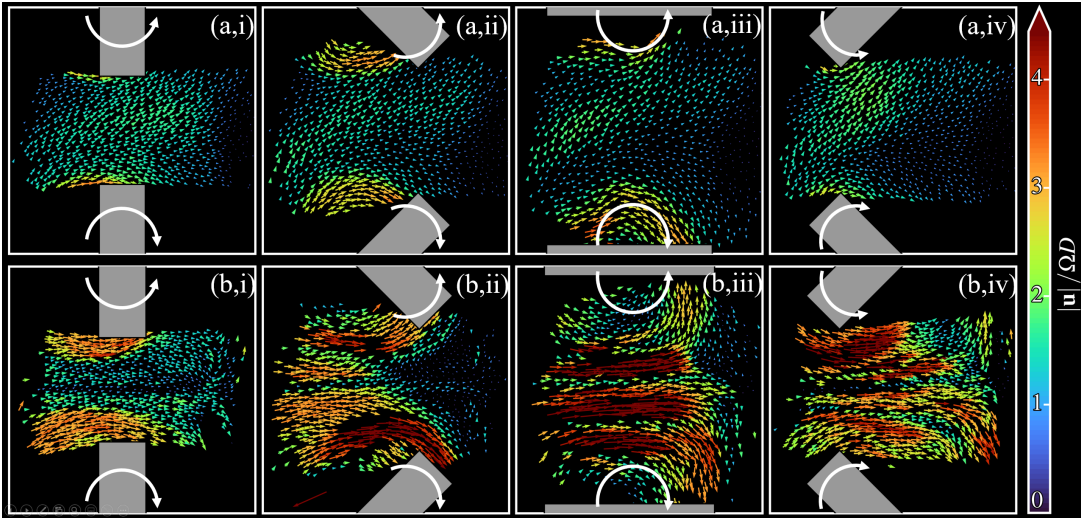


FIG. 11. Top-down visualization of the flow field from Newtonian (a) and 50 ppm (b) cases at rotation rate of 0.125 Hz. For the top and bottom rows, plots (i) through (iv) represent four consecutive snapshots separated by phase difference of $\pi/4$.

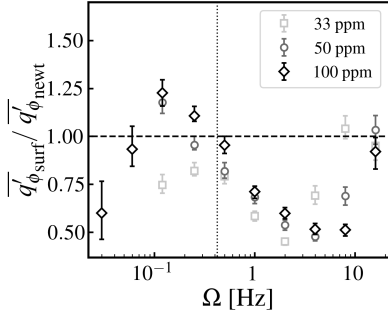


FIG. 12. Ratio of spatially averaged velocity fluctuation RMS (phase-locked) between the surfactant and Newtonian cases.

can be drawn. As the rotation rate of the blades is increased, we reach a point at which the intense shear starts to break the micellar structures which are responsible for the attenuation of turbulence, and the flow state gradually returns to the Newtonian case as the resulting smaller micelles are unable to significantly affect the flow. In the aforementioned channel flow studies, it has also been shown that an increase in the concentration of the solution leads to an increase in the critical Reynolds number at which this breakup happens in the same range

of concentrations as the current experiment. Here, we see that the rotation rate required for the attenuation effect to begin vanishing increases as the concentration of the solution increases, which matches the expected behavior.

Returning to the first trend, below the rotation rate of 0.5 Hz, the results become more complex since the exact behavior depends on the concentration of the solution. Considering the previous analyses, we expect the Kolmogorov timescale to be longer, or slower, than the relaxation time of the surfactant micelles, which would lead to no turbulence modulation and an unchanged level of turbulent intensity. However, as we previously discussed, this region is also marked by the formation of SIS (or streak-like structures) which significantly affect the mean flow. Such larger structures have slower relaxation times than the smaller micellar networks, and thus can interact with the “turbulence” of these slower cases. Combined with the intensification of the mean flow, which is also dependent on the concentration of the solution, the resulting effect of the turbulent intensity in this region becomes more dependent on the specific Reynolds number and solution concentration.

V. CONCLUSIONS

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